CE302 SOFTWARE ENGINEERING PROJECT

Task Manager

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ANALYSIS REPORT

1. Introduction

1.1 Purpose

The purpose of this document is to outline all of the requirements of the TaskFlow — Smart Task Manager Application. The system functionality and interactions are provided below in order to establish an agreement between the developers and stakeholders.

1.2 Background

Due to the rapid changes in our lives, people have trouble balancing their work, study, and private life. The process of keeping track of deadlines, setting priorities and managing progress causes a high level of cognitive load. Traditional note-taking tools lack filtering and searching options, hence making it hard to manage one's goals.

1.3 Motivation

• Statement of problems with the current system: Current manual processes or even basic text editors cannot automate task tracking and sorting by priority and deadlines. People tend to forget about unfinished tasks and progress in general.

• New System: TaskFlow manages to solve all of the above issues as it enables the creation, organization, prioritization, and tracking of tasks in real time. It offers a real-time statistical overview of completed and uncompleted tasks, as well as an advanced filtering feature.

1.4 Structure of the Document

The document is structured as follows: Section 2 discusses the functional requirements, system users, and requirements, which include Use Case models. Section 3 discusses non-functional requirements. Section 4 discusses system models, while definitions and abbreviations can be found in Section 5 and Section 6 respectively.

2. Functional Requirements

2.1 Description of System Functionalities

Given the implementation of the TaskFlow application, the functional requirements are described as follows:

FR1: User Registration/Login

The system will enable the user to register and login so as to view their personal task dashboard using Firebase Authentication.

FR2: Task Creation & Management

The system will allow the user to create a task by filling in the title and description of the task.

FR3: Task Prioritization

The system will allow users to assign priority levels to tasks so as to facilitate sorting of the tasks.

FR4: Category Management

The system will have default categories which are work, personal, education, and health.

The system will also allow users to create their own categories.

FR5: Task Due Date

The system will allow the user to specify a particular due date (date/month/year) for the task.

FR6: Task Search Option

The system will have an option for searching tasks through the title or descriptions of tasks.

FR7: Dashboard Statistics

The system will display dashboard statistics including the total number of tasks, remaining tasks, and completed tasks.

FR8: Filtering and Sorting Options

The system will give options to filter or sort the tasks according to Newest First, Oldest First, Priority Level, and Due Date.

2.2 Description of the System Users

• Standard User: An individual who employs his or her login details for creation, modification, deletion, searching and management of tasks and categories on his or her own.

2.3 Specific Requirements

2.3.1 Use Case List

Among the use cases that have been developed for TaskFlow are:

• UC1: Log In / Sign Up

• UC2: Create Task

• UC3: Set Task Priority

• UC4: Assign Category

• UC5: Create Custom Category

• UC6: Set Deadline

• UC7: Search Tasks

• UC8: View Dashboard Statistics

• UC9: Filter and Sort Tasks

2.3.2 Use Case Priority List

• UC1: Log In / Sign Up

• Priority: High

• Description: Important for data security and dashboard access.

• UC2: Create Task • Priority: High

• Description: Main use case.

• UC6: Set Deadline

• Priority: High

• Description: Important for monitoring the process in real time.

• UC3: Set Task Priority

• Priority: Medium

• Description: Required for tasks organization.

• UC4: Assign Category

• Priority: Medium

• Description: Helpful in assigning categories to tasks.

• UC5: Create Custom Category

• Priority: Medium

• Description: Gives opportunity for customization.

• UC7: Search Tasks

• Priority: Medium

• Description: Makes task management easier when there are lots of tasks.

• UC9: Filtering and Sorting Tasks

• Priority: Medium

• Description: Allows users to concentrate on particular types of tasks.

• UC8: Presentation of Dashboard Statistics

• Priority: Low

• Description: Provides some insight into task management.

2.3.3 Use Case Specification

Use Case UC2: Task Creation

• Actors: Standard User

• Precondition: The user has to log in to the application.

• Normal flow:

1. The user pushes the "New Task" button from the app's main screen.

2. The system shows a form with the following fields: Title, Description, Category, Priority, Deadline.

3. The user provides the name and description for the task.

4. The user sets the category and priority of the task.

5. The user defines the deadline.

6. The user pushes the "Save" button.

7. The system validates and stores the task data.

• Postconditions: The task is created and appears on the user's dashboard.

Use Case UC9: Filtering and Sorting Tasks

• Actors: Standard User

• Precondition: The user is on the main task dashboard and has some tasks there.

• Main flow:

1. User chooses one of the filter/sorting dropdown menus, which is either located in the sidebar or the main panel.

2. User chooses one of the sorting options (Newest, Oldest, Priority, or Due Date).

3. The tasks list will be sorted according to the criteria that user has chosen using Firebase queries.

• Postconditions: Tasks are sorted by the user criteria.

3. Non-Functional Requirements (volere template)

3.1 Usability and humanity requirements

• Requirement #1: Intuitive user interface.

• Description: The system should possess intuitive UI. It allows a user to create a task without any documentation reading.

• Fit criterion: No more than 10% of test users should spend more than 1 minute creating a task.

• Priority: High

3.2 Performance Requirements

• Requirement #2: High performance of tasks.

• Description: The system should execute all operations rapidly. Operations include tasks creating, search, filtering, etc.

• Fit criterion: All operations must be executed not more than in 2 seconds.

• Priority: High

3.3 Security requirements

• Requirement #3: Authentication requirement.

• Description: Access to user dashboards should be provided only to particular users.

• Fit criterion: Login authentication through Firebase auth should be conducted to access tasks.

3.4 Maintainability and Portability Requirements

• Requirement #4: Maintaining and porting code.

• Description: The code must follow the Object Oriented principle very strictly.

• Fit Criterion: The software should be runable across all Web browsers.

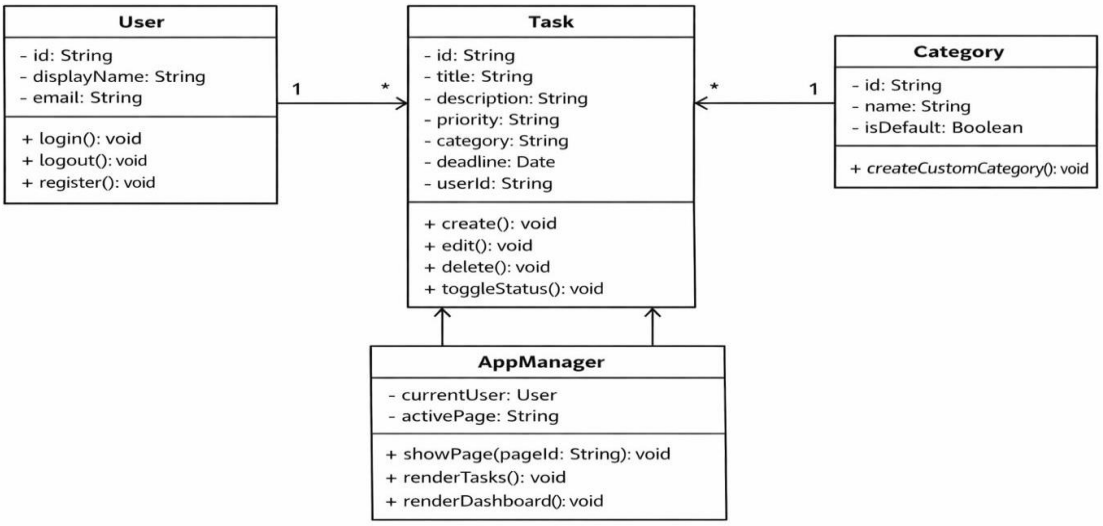
• Priority: Medium

4. System Models

4.1 Object and Class Model

Since the system is designed following Object-Oriented Programming (OOP) principles, the core entities of TaskFlow are modeled as classes. Below is the structured description of the classes, their attributes, and methods to be represented in the Class Diagram.

1. Class: User • Attributes: • - id: String (User's unique Firebase UID) • - displayName: String • - email: String • Methods: • + login(): void • + logout(): void • + register(): void
2. Class: Task • Attributes: • - id: String • - title: String • - description: String • - priority: String • - category: String • - deadline: Date • - userId: String (Foreign key referencing User) • Methods: • + create(): void • + edit(): void • + delete(): void • + toggleStatus(): void
3. Class: Category • Attributes: • - id: String • - name: String • - isDefault: Boolean • Methods: • + createCustomCategory(): void
4. Class: AppManager (Control Class) • Attributes: • - currentUser: User • - activePage: String • Methods: • + showPage(pageId: String): void • + renderTasks(): void • + renderDashboard(): void



4.2 User Interface

The navigational paths and user interface mock-ups for TaskFlow reflect the dynamic single-page structure implemented in the application.

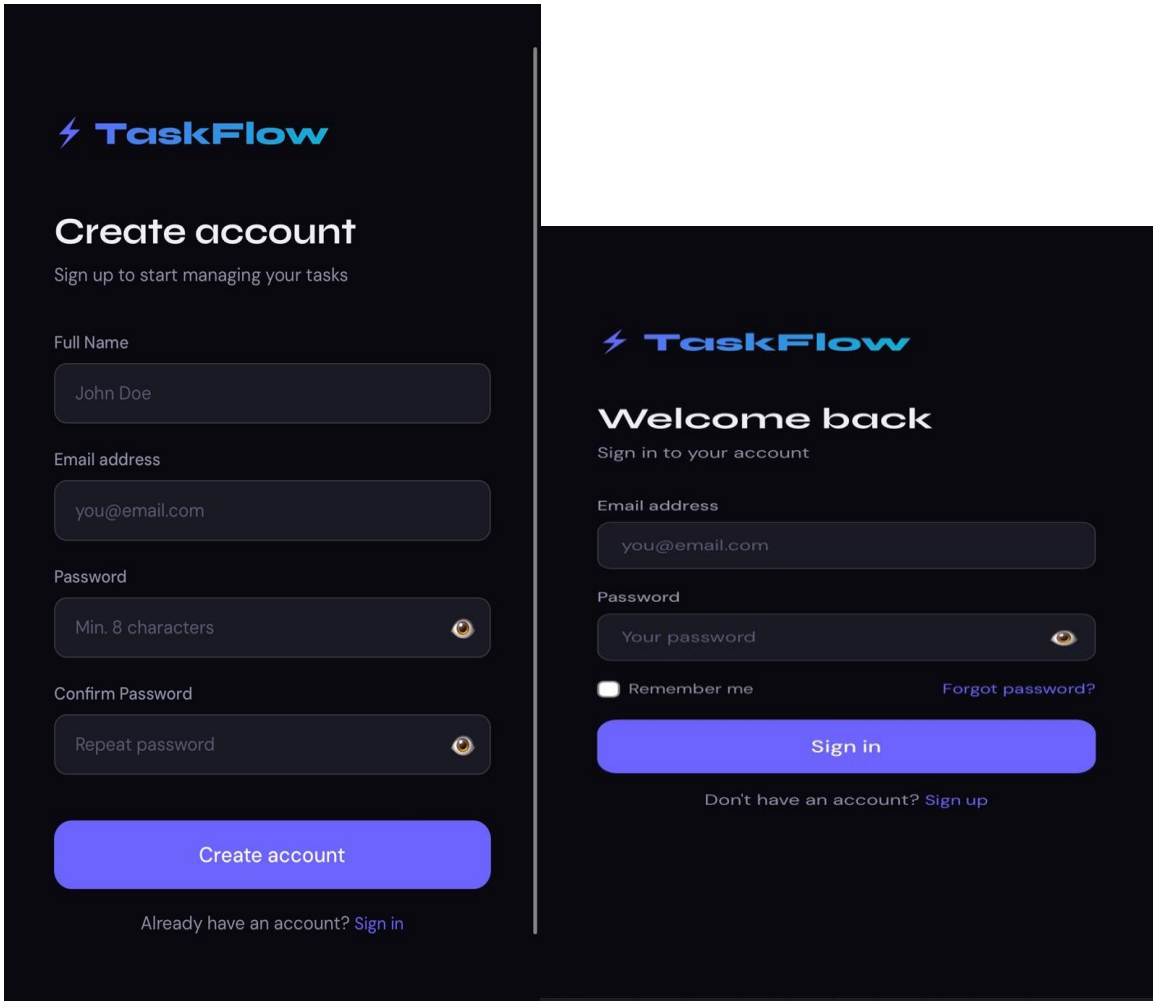
• Navigational Paths:

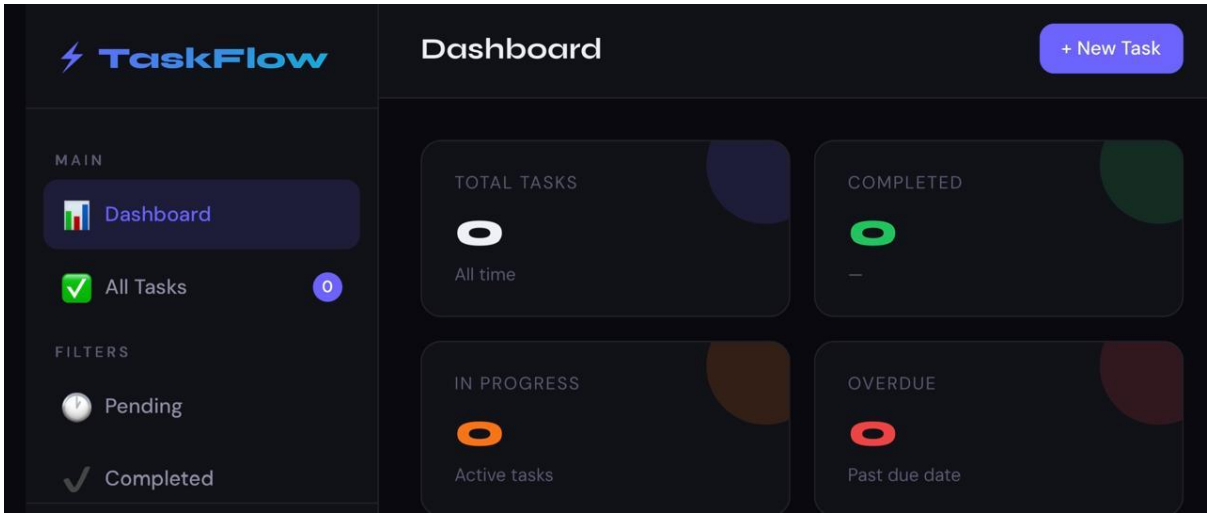
1. Landing Page (page-landing) \rightarrow User selects Sign In or Sign Up.

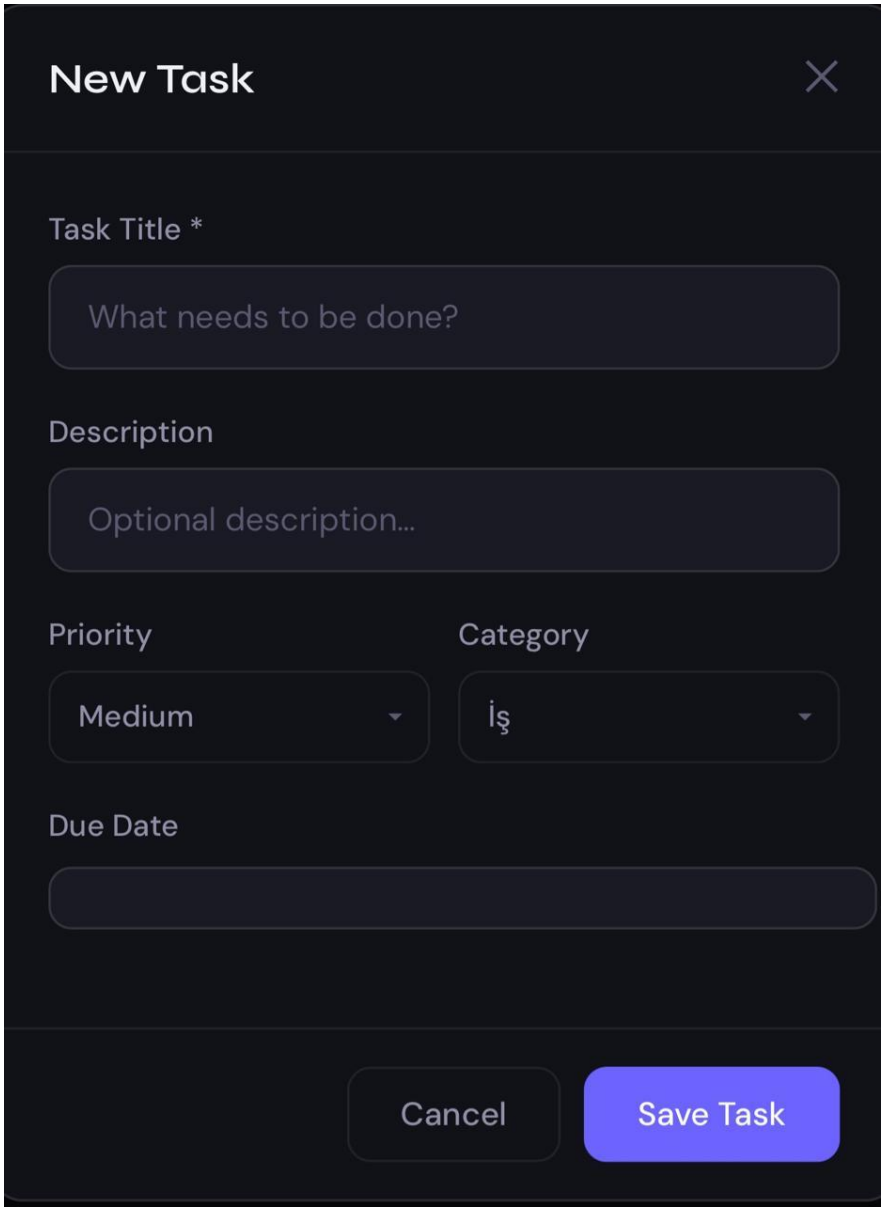
2. Login Page (page-login) \rightarrow User authenticates \rightarrow Redirects to Main App.

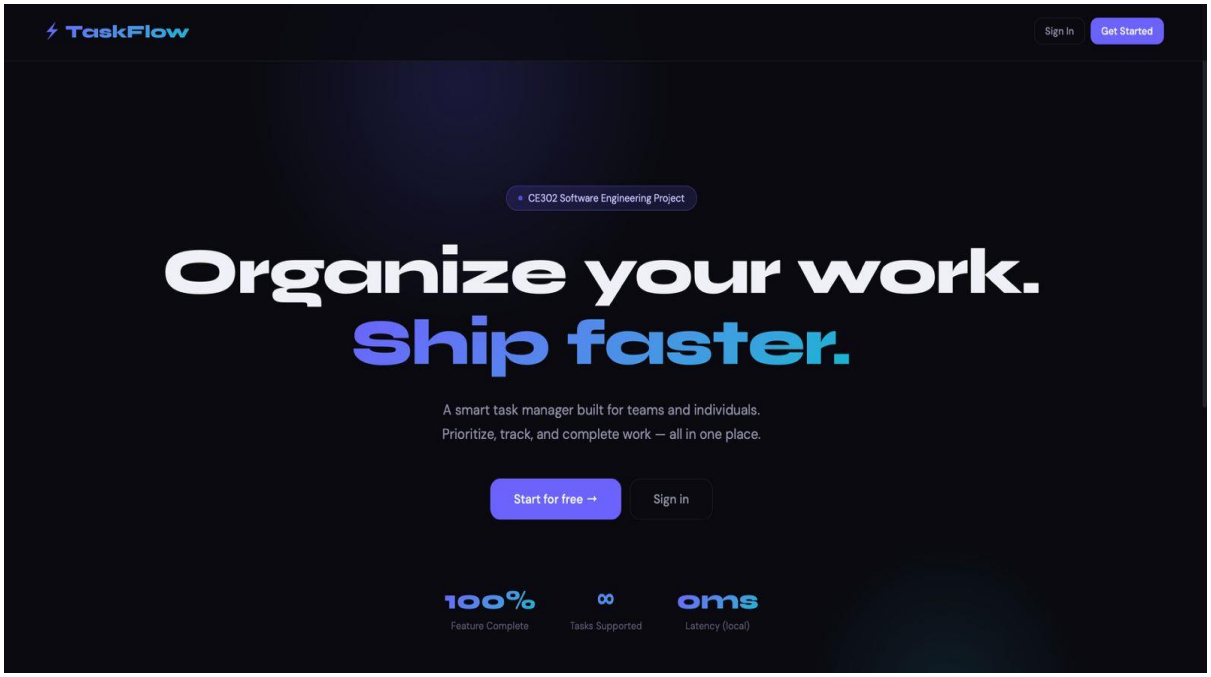
3. Signup Page (page-signup) \rightarrow User registers \rightarrow Redirects to Main App.

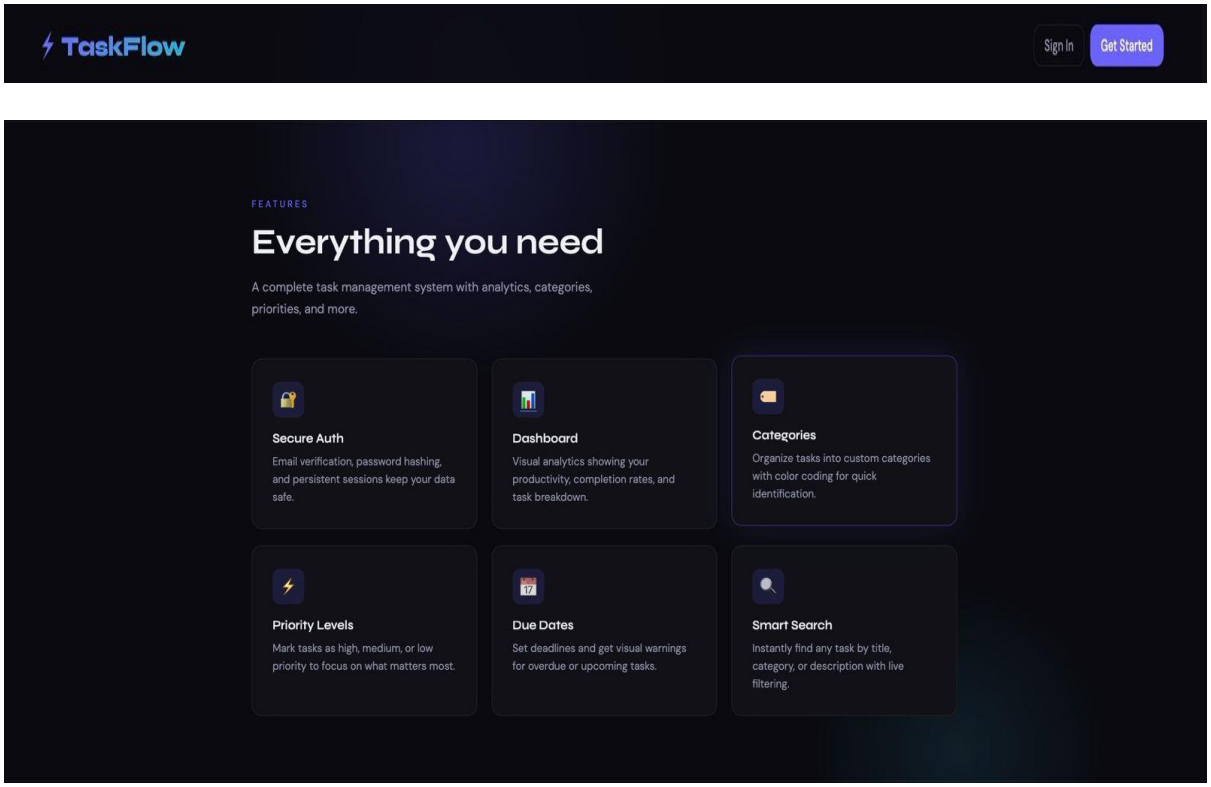
4. Main App Dashboard (page-app) \rightarrow The core environment where tasks are created, filtered, and managed.




TaskFlow
Dashboard
+ New Task
MAIN
Dashboard
All Tasks
0
FILTERS
Pending
Completed
TOTAL TASKS
0
All time
COMPLETED
0
—
IN PROGRESS
0
Active tasks
OVERDUE
0
Past due date


New Task
Task Title *
What needs to be done?
Description
Optional description...
Priority
Medium
Category
İş
Due Date
Cancel
Save Task


TaskFlow
Sign In
Get Started
CE302 Software Engineering Project
Organize your work.
Ship faster.
A smart task manager built for teams and individuals.
Prioritize, track, and complete work — all in one place.
Start for free →
Sign in
100%
Feature Complete
∞
Tasks Supported
0ms
Latency (local)


TaskFlow
Sign In
Get Started
FEATURES
Everything you need
A complete task management system with analytics, categories, priorities, and more.
Secure Auth
Email verification, password hashing, and persistent sessions keep your data safe.
Dashboard
Visual analytics showing your productivity, completion rates, and task breakdown.
Categories
Organize tasks into custom categories with color coding for quick identification.
Priority Levels
Mark tasks as high, medium, or low priority to focus on what matters most.
Due Dates
Set deadlines and get visual warnings for overdue or upcoming tasks.
Smart Search
Instantly find any task by title, category, or description with live filtering.

5. Terms, Acronyms, Abbreviations

Here below the list of all definitions for terms, acronyms, and abbreviations used in the report is provided to maintain consistency in the understanding of all readers.

- Firebase – an application development platform by Google which offers various services such as databases, analytics, and authentication.
- Firestore – cloud database offered by Firebase; NoSQL database used in the TaskFlow application.
- FR – Functional Requirement.
- NFR – Non-Functional Requirement.
- OOP – Object-Oriented Programming; programming paradigm that centers on objects that can contain both data and code.
- TaskFlow – name for the application that we plan to build.
- UC – Use Case.
- UI/UX – User Interface/User Experience.

6. Glossary & References

6.1 Glossary

• Actor – user role or role of the external system while working with TaskFlow application.

• Mock-Up – a static visual representation of the application interface showing design features.

• Single-Page Application (SPA) – a website/application created as a single-page site based on dynamic reloading technique rather than loading the whole webpage.

• Volere Template – template for describing requirements, especially non-functional ones.

6.2 References

[1] Google, "Firebase Documentation: Web Setup and Firebase Authentication," Google Firebase.

[2] Google, "Cloud Firestore Documentation: Get Data and Realtime Updates," Google Firebase.

[3] MDN Web Docs, "JavaScript Modules (ESM) in the Browser," MDN.

[4] Atlantic Systems Guild, "Volere Requirements Specification Template," Volere.

[5] Course Instructor. Advanced Software Engineering: Course Project Guidelines and Requirements Specification. [Internal Document].